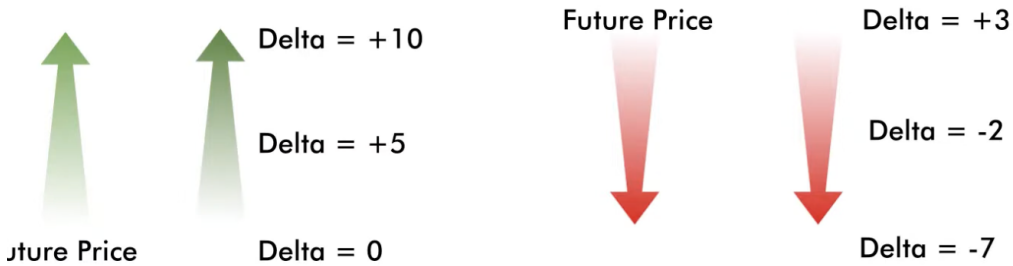

Figure 37: Gamma changes delta in the same direction as the underlying move. As the market rises, delta becomes more positive; as it falls, delta becomes more negative.

Because long gamma is usually paid for through **theta**.

As time passes, options lose value all else equal. That time decay is the cost of carrying long gamma.

So the real economic tradeoff is:

- a **long gamma** portfolio tends to benefit from movement,
- a **short gamma** portfolio tends to benefit from time decay.

Remark 16.11. A long-gamma position makes money when the realized movement of the underlying is large enough to outweigh the theta paid. A short-gamma position makes money when the theta collected is large enough to outweigh the losses from re-hedging adverse movement.

This gamma-theta tradeoff is one of the central tensions in options market making and will appear repeatedly throughout the rest of these notes.

Remark 16.12. Gamma is what turns delta from a static number into a dynamic risk. Once the market moves, gamma is the reason the hedge must move too.

17 Theta

17.1 Definition and Core Intuition

Definition 17.1 (Theta)**.** **Theta**, denoted by Θ, is the amount by which an option's theoretical value decreases as time passes, usually measured over the next 24-hour period.

Theta is the Greek of *time decay*. If all other inputs are held fixed, then an option becomes worth less as expiration gets closer.

Remark 17.1. Traders usually describe theta in terms of **paying theta** or **collecting theta**. If you are long options, you pay theta. If you are short options, you collect theta.

Remark 17.2. For an individual long option, theta is negative because time passing hurts the owner of optionality. From the opposite side, a short option position benefits from that same decay.

17.2 A Basic Example

Suppose we are long 100 options priced at 2.00, each with multiplier 100. If after 24 hours the model reprices them at 1.93, then the theta per option is

$$\Theta = -0.07.$$

So the total theta of the position is

$$-0.07 \times 100 \times 100 = -700.$$

This means the trader expects to lose about $700 from time decay over the next day, assuming all other pricing inputs remain unchanged.

Example 17.1 (Theta from a quoted option)**.** Suppose a call has theoretical value

$$11.04$$

and theta

$$-0.12.$$

Then, holding everything else constant, the model says that in 24 hours the option should be worth about

$$11.04 - 0.12 = 10.92.$$

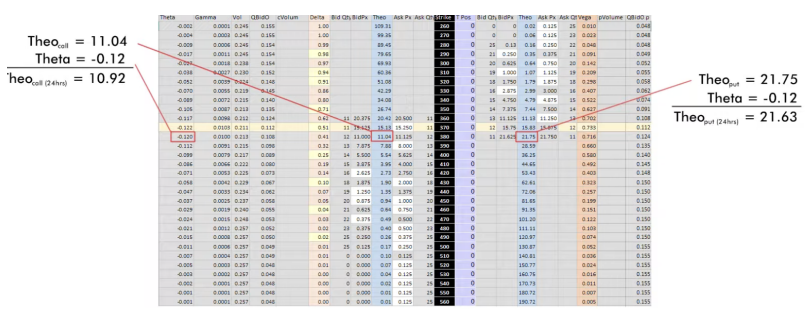

Figure 38: Example of theta on an options board. The call and put at the same strike lose the same amount of time value over the next 24 hours.

17.3 Theta for Calls and Puts at the Same Strike

One important relationship is that a call and a put with the same strike and expiration have the same theta contribution from time decay, provided we are holding the other pricing inputs fixed.

This is consistent with put-call parity. If the time decay of same-strike calls and puts were materially inconsistent, the relationship between call value, put value, strike, and underlying would not line up properly.

Remark 17.3. So when looking at a board, it is often useful to remember that the call and put on the same strike are tied together not only through parity, but also through the time-decay logic of the pricing model.

17.4 Theta as a Known Cost

Theta is especially useful because it represents a relatively predictable drag on option value over a short horizon.

If you are long options, theta is the known cost of carrying that convexity for another day. The uncertain question is whether the market will move enough for your gamma or directional exposure to overcome that cost.

Remark 17.4. This is one of the most practical ways to think about theta: it is the bill you pay for continuing to own optionality.

17.5 Theta Versus Gamma

The most important practical relationship involving theta is the tradeoff between **theta paid** and **gamma gained**.

A long-option position is long gamma, which means it can profit from underlying movement through re-hedging. But that same position also pays theta every day.

So the real question is:

Did the realized movement of the underlying outweigh the theta decay paid?

Example 17.2 (Gamma scalp versus theta decay)**.** Suppose a trader owns 100 options with multiplier 100, and the theta is

$$-0.12.$$

Then the 24-hour theta cost is

$$-0.12 \times 100 \times 100 = -1200.$$

Now suppose the trader's gamma hedging over that same period generates

$$+600$$

of hedge profit.

Then the total result is

$$-1200 + 600 = -600.$$

So even though gamma scalping helped, the realized movement was not large enough to offset the theta paid.

Remark 17.5. This comparison is one of the central ideas in options trading. Long gamma is valuable only if the movement you harvest is large enough to justify the theta you pay.

Another way to say this is that if the option buyer makes more from movement than they lose from decay, then the underlying *outrealized* what the option's implied pricing had embedded for that period.

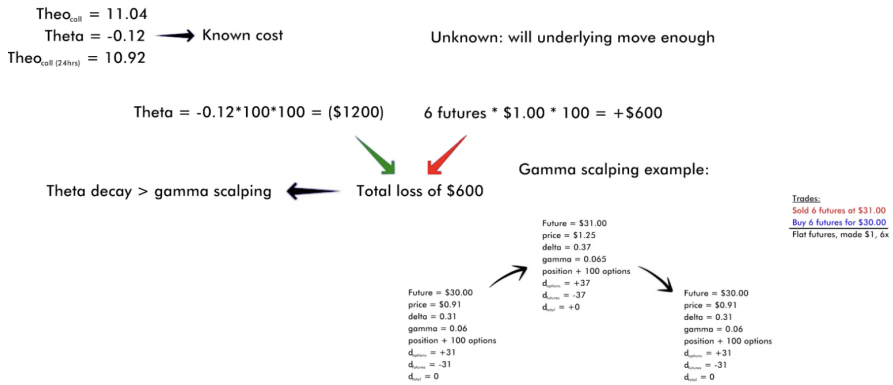

Figure 39: Theta versus gamma example. The known decay cost from theta is compared against the uncertain hedge profit generated from gamma scalping.

17.6 Theta Across Strike Space

Theta is not uniform across strikes. It is generally largest in magnitude near the at-the-money region.

That happens because ATM options still have the most time value to lose, and they are most sensitive to the narrowing window before expiration.

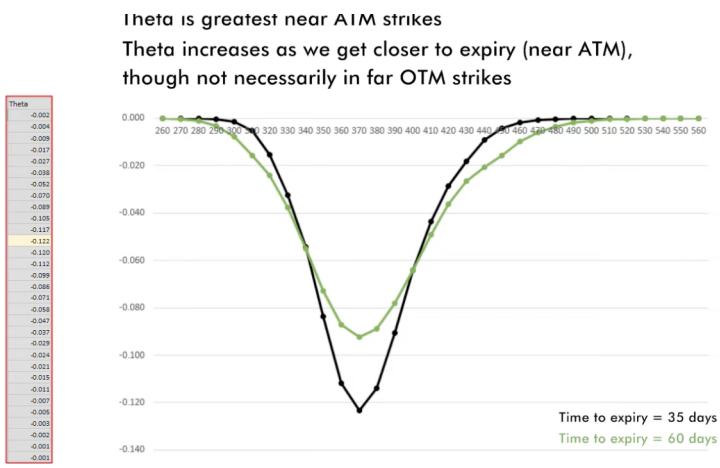

Figure 40: Theta is greatest near at-the-money strikes. The near-expiry curve shows stronger time decay around ATM than the longer-dated curve.

Remark 17.6. Like gamma, theta tends to concentrate around ATM options. But unlike gamma, theta is a decay Greek, so the practical focus is usually on how expensive that optionality is to continue holding.

17.7 Theta Across Time to Expiration

Theta also changes with time to expiration. For at-the-money options, theta usually becomes larger in magnitude as expiration gets closer.

That means the rate of decay accelerates as maturity gets short, especially for ATM strikes.

For out-of-the-money strikes, the pattern can be less clean. Some far OTM options may show relatively small theta near expiry because there is already very little premium left to lose.

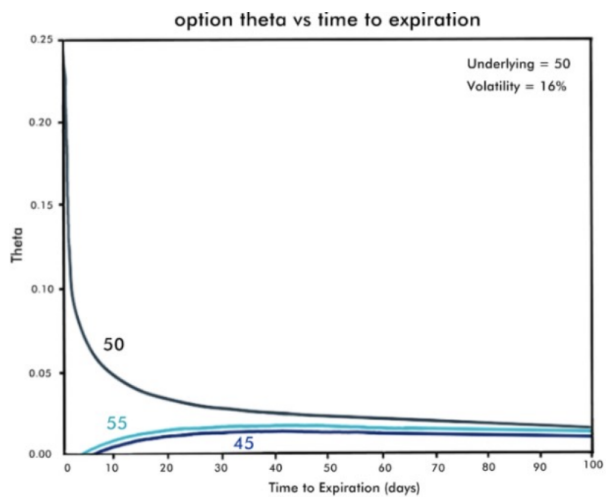

Figure 41: Theta across time to expiration for different strikes. At-the-money options show the strongest increase in decay as expiry approaches.

Remark 17.7. The key takeaway is not that every strike behaves identically, but that for strikes near the money, theta tends to grow in magnitude as expiration approaches.

17.8 Theta Surface

We can also view theta as a surface across strike and maturity. This shows that the strongest decay is concentrated near the ATM region, with the effect becoming sharper as expiration shortens.

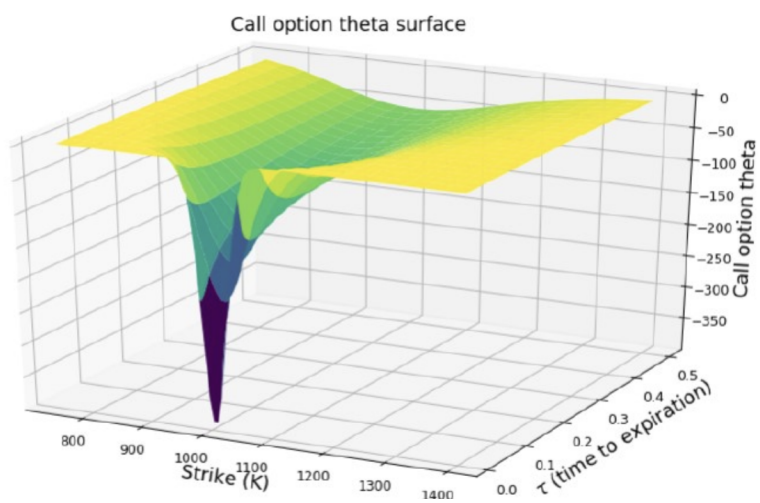

Figure 42: Call option theta surface across strike and time to expiration. The deepest negative theta appears near at-the-money strikes and short maturities.

Remark 17.8. The theta surface resembles the gamma surface in where it concentrates: near the at-the-money region and near expiration. This is not an accident. Gamma and theta are tightly linked through the economics of owning convexity.

17.9 A Practical Theta Rule

From a trader's point of view, theta can be summarized in a simple way: